

2013-2014 – Master 2 – Macro I

Lecture 2 : Balance Growth Paths

Franck Portier

(based on Gilles Saint-Paul lecture notes)

franck.portier@TSE-fr.eu

Toulouse School of Economics

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Disclaimer

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1. What is a balanced growth path ?

Growth rates

- ▶ In a continuous time model, the growth rate of a variable X is given by

$$g_X(t) = \frac{1}{X(t)} \frac{dX(t)}{dt} = \frac{d}{dt} \ln X(t).$$

- ▶ In a discrete time model, the growth rate of X is given by

$$g_X(t) = \frac{X(t) - X(t-1)}{X(t-1)}.$$

- ▶ If the growth rate is small, then also in this case :

$$g_X(t) \approx \ln \frac{X(t)}{X(t-1)}.$$

- ▶ In what follows I will assume time is continuous.

1. What is a balanced growth path ?

Definition

- **Definition 1 :** Consider any dynamic model. Then a *balanced growth path* of that model is a trajectory such that all variables grow at a constant rate. That is

$$\dot{X}(t) = g_X X(t), \text{ i.e. } X(t) = X(0)e^{g_X t}$$

- **Remark 1 :** Note that this does *not* mean that all variables grow at the same rate. However, variables will typically be sorted in groups such that all variables in the same group grow at the same rate.

1. What is a balanced growth path ?

Theorem 1

- **Theorem 1 :** Let Y and $X_1...X_n$ be strictly positive variables such that

$$Y(t) = \sum_{i=1}^n a_i X_i(t),$$

holds, for all t , with $a_i > 0$. Then in a balanced growth path $Y, X_1, ..., X_n$ all grow at the same rate.

- **Proof :** Take all the variables that grow at a maximum rate on the RHS. This must also be the rate at which $Y(t)$ grows, otherwise one side would become negligible compared to the other asymptotically, and the two sides could not remain equal. Take all those variables and subtract them from both sides. Assume some variables remain on the RHS, meaning the RHS (and therefore the LHS is strictly positive. We then get an equality where the LHS is strictly positive and grows at a greater rate than all the terms on the RHS. This is clearly impossible, so it must be that all variables on the RHS grow at that maximum rate. Therefore, all variables grow at the same rate.

1. What is a balanced growth path ?

Corollary

- ▶ Thus, for any such *additive* relationship, we know that all the variables involved grow at the same rate.
- ▶ The same is true if the relationship involves any time derivative of any variable, since those derivatives grow at a constant rate which is the same as that of the original variable.
- ▶ Thus it is straightforward to prove the following corollary :
- ▶ **Corollary 1 :** Let Y and $X_1, \dots, X_n$ be strictly positive variables such that

$$Y(t) = \sum_{i=1}^n a_i X_i(t) + \sum_{i=1}^n b_i \frac{dX_i(t)}{dt}, \quad (1)$$

with $a_i \geq 0$, $b_i \geq 0$ and at least one a_i or b_i strictly positive for each i . Then in a balanced growth path such as $Y, X_1, \dots, X_n$ all grow at a strictly positive rate, they grow at the same rate.

1. What is a balanced growth path ?

Remarks

- ▶ **Remark 1 :** Note that we need to impose the restriction that the growth rate of the variables involved is positive. That is because if a variable grows at a zero rate its derivative is equal to zero and disappears from the relationship. For example if $Y = X_1 + \dot{X}_2$ we may have $g_{X_2} = 0$ and $g_Y = g_{X_1} > 0$. That is, to apply the theorem all the RHS variables in (1); must be strictly positive, implying that the X_i s must grow at a positive rate.
- ▶ **Remark 2 :** A counter example to Theorem 1 is easy to come with if we relax the assumption that the X s and the a s are strictly positive. For example, if we have $Y = X_1 + X_2 - X_3$, one may have a BGP where $Y = X_1$ and $X_2 = X_3$, and $g_Y = g_{X_1} \neq g_{X_2} = g_{X_3}$.

1. What is a balanced growth path ?

Steady State and BGP existence

- ▶ Note that a balanced growth path should not be confused with a steady state.
- ▶ **Definition 2 :** A *steady state* is a trajectory where all variables are constant.
- ▶ **Existence :** There is no general presumption that a balanced growth path or a steady state exists in a given model.

2. BGPs in neo-classical models

A neo-classical model

- ▶ A central question of macroeconomic theory is : how can an economy grow in the long-run ? In general, one typically assumes that in the long run, growth has to be "balanced". In fact, there is no compelling reason to believe this empirically, other than the somewhat fuzzy "stylized facts of growth".
- ▶ So the key question is : how can an economy grow ?
- ▶ To address this question, it is convenient to consider a "Neo-classical" economy, whose key ingredients are :
 - ▶ A production function

$$Y(t) = F(K(t), L(t)) \quad (2)$$

- ▶ An equation for capital accumulation

$$\dot{K}(t) = Y(t) - C(t). \quad (3)$$

where Y = output, L = labour, K = capital, and C = consumption.

- ▶ $\forall t$ $L(t)$ is equal to the workforce which is an exogenous variable. I will assume it grows at a constant rate g_L .

2. BGPs in neo-classical models

BGP

- ▶ Assume $g_K > 0$.
- ▶ Since $Y = C + \dot{K}$, we know from Corollary 1 that $g_K = g_C = g_Y = g$.
- ▶ Log-differentiating (2), we have that

$$g_Y = \eta_K g_K + \eta_L g_L,$$

where

$$\eta_K = \frac{K}{Y} \frac{\partial F}{\partial K}$$

is the elasticity of production with respect to capital, and similarly for η_L .

- ▶ Important remark : η_K should be read as $\eta_K(t)$, as it needs not to be constant (same thing doe η_L)
- ▶ Thus along any BGP, it must be that

$$(1 - \eta_K)g = \eta_L g_L.$$

2. BGPs in neo-classical models

BGP (continued)

$$(1 - \eta_K)g = \eta_L g_L.$$

- ▶ Whether this can be fulfilled crucially depends on the production function.
- ▶ In a neo-classical case, there are constant returns to scale to all factors of production. By Euler's theorem, this means that $\eta_K + \eta_L = 1$.
- ▶ The previous equation then implies that $g = g_L$.
- ▶ Thus the growth rate of output in a BGP must be equal to that of labor. In particular, output per capita Y/L remains constant : there can be no growth in per capita terms.

2. BGPs in neo-classical models

Remarks

- ▶ **Remark 4 :** No assumption on savings behaviour is needed to derive this result.
- ▶ **Remark 5 :** One can have $g > 0$ and $g_L = 0$ in the extreme case where $\eta_K = 1$, i.e. only capital enters the production function.
- ▶ **Remark 6 :** Under constant returns the production function can be rewritten $Y(t) = L(t)f(\frac{K(t)}{L(t)})$. The model can then be re-cast in terms of the per-capita variables, $y(t) = Y(t)/L(t)$, $k(t) = K(t)/L(t)$, and $c(t) = C(t)/L(t)$. Then the model's equations are

$$\begin{aligned}y(t) &= f(k(t)) \\ \dot{k}(t) &= y(t) - c(t) - g_L k(t).\end{aligned}$$

2. BGPs in neo-classical models

The five stylized fact for growth

SF1 : Y/L and K/L go up over time

SF2 : K/Y is constant

SF3 : Wages go up

SF4 : The profit rate is constant

SF5 : The share of labour and capital in GDP are constant.

2. BGPs in neo-classical models

The need for technical progress

- ▶ The stylized facts of growth tell us that output per capita grows in the long-run. This is not compatible with the preceding model.
- ▶ Why? Because there are decreasing returns to capital. To sustain growth, $\dot{K}$ must keep up with K . But $\dot{K}$ (=investment) comes from savings, which cannot exceed output.
- ▶ Unfortunately, each additional unit of capital (per capita) delivers less and less output (per capita) because of decreasing returns. This prevents growth in output per capita from being sustained in the long run.
- ▶ The way to get growth per capita is to add an exogenous trend of technical progress, called $A(t)$.
- ▶ One could represent that by assuming a general production function $F(K, L, A)$ but let us focus on the three standard ways.

2. BGPs in neo-classical models

Labour-augmenting technical progress

- ▶ We have

$$Y(t) = F(K(t), A(t)L(t)).$$

- ▶ Under constant returns this can be rewritten :

$$Y(t) = A(t)L(t)f\left(\frac{K(t)}{A(t)L(t)}\right).$$

- ▶ Therefore in per-capita terms :

$$y(t) = A(t)f\left(\frac{k(t)}{A(t)}\right),$$

with $y = Y/L$ and $k = K/L$.

- ▶ Differentiating

$$g_y = g_A + \eta_K(g_k - g_A). \quad (4)$$

- ▶ Denote g the common growth rate of output, capital and consumption, we have $g = g_A$.
- ▶ the economy grows at the same rate as the rate of technical progress.

2. BGPs in neo-classical models

Labour-augmenting technical progress (continued)

- ▶ The elasticity that appears in equation (4) should be equal to $\frac{k/A \cdot f'(k/A)}{f(k/A)}$, but we can check that this is also equal to η_K as defined above.
- ▶ Indeed

$$\eta_K = \frac{K}{Y} \frac{\partial Y}{\partial K} = \frac{K}{ALf()} \frac{ALf'()} {AL} = k/A \cdot f'()/f().$$

2. BGPs in neo-classical models

Capital-augmenting technical progress

- ▶ There we have

$$Y(t) = F(A(t)K(t), L(t)),$$

which we will now rewrite as $Y(t) = L(t)f\left(\frac{A(t)K(t)}{L(t)}\right)$.

- ▶ The per-capita production function is then

$$y(t) = f(A(t)k(t)).$$

- ▶ Along a BGP we must have

$$g_y = \eta_K(g_A + g_k).$$

- ▶ Therefore one must have $g = \frac{\eta_K}{1-\eta_K}g_A$.
- ▶ For a BGP to exist the elasticity η_K must be constant along the BGP.
- ▶ This will not be in general true unless the production function is Cobb-Douglas : $Y = BK^\alpha L^{1-\alpha}$. In this case $\eta_K = \alpha$.

2. BGPs in neo-classical models

Output-augmenting technical progress

- ▶ This is defined by

$$Y(t) = A(t)F(K(t), L(t)).$$

- ▶ We now have $y(t) = A(t)f(k(t))$, and $g_y = g_A + \eta_K g_k$.
- ▶ Now we must have $g = \frac{g_A}{1-\eta_k}$.
- ▶ There is no BGP unless one is in the Cobb-Douglas case.
- ▶ **Remark 8 :** In the Cobb-Douglas case the three formulations are equivalent up to a redefinition of $A(t)$.

3. Balanced growth and the “stylized facts”

Setup

- ▶ We are going to show that any neo-classical balanced growth path with $g_y > 0$ satisfies the 5 stylized facts.
- ▶ We know that in such a BGP, y , c , and k grow at the same rate; that this rate is equal to the rate of technical progress; and that technical progress must essentially be labour-augmenting.
- ▶ The production function is

$$Y(t) = F(K(t), A(t)L(t)) = A(t)L(t)f\left(\frac{K(t)}{A(t)L(t)}\right).$$

3. Balanced growth and the “stylized facts”

Checking the stylized facts

- ▶ **SF1** : Indeed $y = Y/L$ and $k = K/L$ both grow at rate g
- ▶ **SF2** : Since K and Y grow at the same rate, K/Y is constant
- ▶ **SF3** : If firms are wage-takers on the labor market, the wage at date t is equal to

$$\begin{aligned}w &= \frac{\partial}{\partial L} F(K(t), A(t)L(t)) \\&= A(t) \left(f\left(\frac{k(t)}{A(t)}\right) - \frac{k(t)}{A(t)} f'\left(\frac{k(t)}{A(t)}\right) \right).\end{aligned}$$

Since $k(t)$ and $A(t)$ grow at the same rate, the term in brackets is constant. It follows that w is also proportional to $A()$ and grows at rate g .

3. Balanced growth and the “stylized facts”

Checking the stylized facts (continued)

- SF4 : Now let us compute the profit rate :

$$\begin{aligned} r &= \frac{\partial}{\partial K} F(K(t), A(t)L(t)) \\ &= f' \left(\frac{k(t)}{A(t)} \right) \\ &= \text{Constant.} \end{aligned}$$

- SF5 : Let us compute the capital share, called s_K :

$$s_K = \frac{rK}{Y} = \frac{f' \left(\frac{k(t)}{A(t)} \right) K(t)}{A(t)L(t)f \left(\frac{K(t)}{A(t)L(t)} \right)} = \frac{k(t)}{A(t)} \frac{f' \left(\frac{k(t)}{A(t)} \right)}{f \left(\frac{k(t)}{A(t)} \right)} = \eta_K.$$

Since k/A is constant along a BGP, this is indeed constant. The same holds for $s_L = 1 - \eta_K = \eta_L$. Note that this constancy of factor shares along a BGP does *not* rest on a Cobb-Douglas assumption.

3. Balanced growth and the “stylized facts”

Remaks

- ▶ **Remark :** The 5 stylized facts of growth must hold in any balanced growth path of a neo-classical model, regardless of what is assumed about preferences.
- ▶ **Remark :** Do not confuse the fact that η_K is constant *along a BGP* regardless of the production function as long as technical progress *is* labor-augmenting, with the fact that for it to remain constant when technical progress *is not* labor-augmenting, the production function must be Cobb-Douglas. In the first case, η_K is constant because it is a function of k/A , which must be constant along a BGP. On the other hand if, for example, technical progress is capital augmenting, then η_K is a function of Ak , which grows with time along a BGP. Thus for η_K to be constant along a BGP it must also be a constant as a function of its argument.